

APPLICATIONS OF QUADRIVARIATE EXPONENTIAL DISTRIBUTION TO A THREE SERVICE POINT TANDEM QUEUE

V. S. Vaidyanathan

Department of Statistics, Pondicherry University, Puducherry-605014, India

and

V. S. S. Yadavalli¹

Department of Industrial and Systems Engineering, University of Pretoria, Pretoria 0002, South Africa

e-mail: *sarma.yadavalli@up.ac.za*

and

P. Chandrasekhar

Department of Statistics, Loyola College, Chennai-600 034, India

and

S. Xavier

Department of Statistics, Loyola College, Chennai-600 034, India

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Summary: A single channel three service point tandem queue with zero queue capacity and blocking is considered. Assuming that the interarrival time and the service times at three service points are dependent, the transient solution for the system, when the joint distribution of interarrival time and service times at service point 1, service point 2 and service point 3 is quadrivariate exponential is obtained. Also, CAN estimator and asymptotic confidence limits for the expected number of entities in the system are obtained.

1. Introduction

Most of the studies on several queueing models are confined to only obtaining expressions for transient or stationary (steady state) solutions and do not consider the associated statistical inference problems. Nowadays, statistical analysis of queueing systems is an important area of research in queueing theory. Parametric estimation, interval estimation and Bayesian inference are some of the statistical tools to understand any random phenomena using stochastic models. Analysis of queueing systems in this direction has not received due attention in the past. Whenever the systems are fully observable in terms of their random components such as interarrival times and service times,

¹ Corresponding author.

standard parametric techniques of statistical theory are quite appropriate. Recently, Bhat (2003) has provided an overview of methods available for estimation, when the information is restricted to the number of customers in the system at some discrete points in time. Bhat (2003) has also described how maximum likelihood estimation is applied directly to the underlying Markov chain in the queue length process in M/G/1 and GI/M/1 queues. Table 1 indicates the present state of work of queueing systems, wherein the asymptotic confidence limits for measures of system performance are obtained.

Table 1: Present state of work of queueing systems.

System Description	Authors	Confidence limits obtained for
M/M/1/ ∞ and M/M/1/N	Yadavalli et al. (2004)	W_Q
M/M/c/ ∞ and M/M/c/N	Yadavalli et al. (2006)	W_Q
Tandem queue with blocking and dependent structure for service times	Chandrasekhar et al. (2006)	L_s, W_s
M/M/R queue with heterogeneous servers	Wang et al. (2006)	L_s
Tandem queue with blocking	Chandrasekhar et al. (2008)	W_s
Bulk-arrival queueing system $M^{[X]}/M/1$	Savariappan et al. (2009b)	L_s
Tandem and bulk service queueing systems	Savariappan et al. (2009a)	L_s
Two station and three station tandem queues with blocking	Chandrasekhar and Vaidyanathan (2009)	L_s
Tandem queue with blocking and dependent structure for interarrival and service times	Chandrasekar et al. (2011)	L_s, W_s

Generally speaking, the queueing models assume that each service channel consists of only one service point. Situations do exist where each service channel may consist of several service points in series. In this situation, a customer must pass successively through all these service points before completing his service. Such situations are known as queues in series or tandem queues. e.g. (a) In a manufacturing process, units must pass through a series of service channels (work stations), where each service channel performs a given task or job, (b) In a university registration process, each registrant must pass through a series of counters such as advisor, department chairman, cashier etc., (c) In a clinical physical examination procedure, a patient must go through a series of stages

such as laboratory tests namely ECG, chest X-ray etc. In all these models, it is not only sufficient to know how many persons are there in the system but also where they are.

In general, the interarrival time and service time are independent random variables and hence there is a necessity to study a model by relaxing this imposed condition. Thus there is a need for a bivariate or a multivariate exponential distribution both from theoretical and simulation studies point of view in Queueing Theory context. In the past, a number of bivariate exponential distributions have been proposed and studied in the literature to describe the service times of two service point tandem queues. But, the bivariate exponential distribution of Marshall and Olkin (1967) is widely accepted among many bivariate exponential distributions in statistical literature mainly due to the fact that it satisfies lack of memory property. Chandrasekhar et al. (2006) have obtained the measures of system effectiveness for a two service point tandem queue, wherein the arrivals follow Poisson process and the service times at service point 1 and service point 2 are governed by bivariate exponential distribution due to Marshall and Olkin (1967). Chandrasekar et al. (2011) have also studied in detail a two service point tandem queue, wherein the interarrival time and service times at service point 1 and service point 2 follow trivariate exponential distribution.

An attempt is made in this paper to extend these results to a three service point tandem queue under the assumption that the interarrival and service times at three service points are dependent and the underlying distribution is quadrivariate exponential. Further, both transient and steady state solutions for the single channel three service points tandem queue with zero capacity and blocking, CAN estimator and asymptotic confidence limits for the expected number of entities in the system are obtained. In the next section, the system description and assumptions are explained in detail.

2. Model: Three service point tandem queue with dependent interarrival and service times

2.1. System description and assumptions

Consider a simplified one-channel queueing system consisting of three service points arranged in series as below.

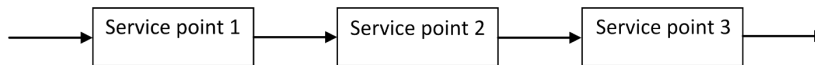


Figure 1: System configuration.

An entity arriving for service must pass through service point 1, service point 2 and service point 3 before completing its service. The assumptions of the model are as follows:

- a. Let T_1 , T_2 and T_3 denote the service times of service point 1, service point 2 and service point 3 respectively and R denote the interarrival time. Assuming that the three service points are identical, it is only appropriate to consider the following quadrivariate exponential distribution

for T_1, T_2, T_3 and R with the survival function given by

$$\bar{F}(t_1, t_2, t_3, t_4) = e^{-[\lambda_1(t_1+t_2+t_3)+\lambda_2(t_1+t_2+t_3)+\lambda_3t_4+\lambda_4\{\max(t_1,t_4)+\max(t_2,t_4)+\max(t_3,t_4)\}]}, \quad (1)$$

$$t_1, t_2, t_3, t_4 \geq 0; \lambda_1, \lambda_2, \lambda_3 > 0; \lambda_4 \geq 0$$

and is denoted by $(T_1, T_2, T_3, R) \sim QVE(\lambda_1 + \lambda_2, \lambda_1 + \lambda_2, \lambda_1 + \lambda_2, \lambda_3, \lambda_4)$, see Marshall and Olkin (1967).

- b. No queue is allowed in front of service point 1, service point 2 and service point 3.
- c. Service point i is said to be blocked, if the entity in service point i completes its service before service point $(i + 1)$ becomes free, $i = 1, 2$.
- d. At any given instant of time, service point 1 is either free or busy or blocked; service point 2 is either busy or blocked, whereas service point 3 is busy.

Note: It may be observed that the following results (i.) - (vii.) are directly obtained from the survival function given in (1), see Marshall and Olkin (1967).

- i. The service times T_1, T_2 and T_3 of service point 1, service point 2 and service point 3 respectively are independent and identically distributed (i.i.d.) exponential random variables each with the parameter $(\lambda_1 + \lambda_2 + \lambda_4)$.
- ii. The interarrival time R is exponential with the parameter $(\lambda_3 + 3\lambda_4)$.
- iii. The joint distribution of (T_i, R) is bivariate exponential (BVE) each with the parameters $(\lambda_1 + \lambda_2, \lambda_3 + 2\lambda_4, \lambda_4)$, $i = 1, 2, 3$.
- iv. The joint distribution of (T_i, T_j, R) is trivariate exponential each with the parameters $(\lambda_1 + \lambda_2, \lambda_1 + \lambda_2, \lambda_3 + \lambda_4, \lambda_4)$, $i, j = 1, 2, 3; i \neq j$.
- v. The quadrivariate exponential distribution satisfies lack of memory property (LMP).

$$\text{i.e., } \bar{F}(s_1 + t, s_2 + t, s_3 + t, s_4 + t) = \bar{F}(s_1, s_2, s_3, s_4) \cdot \bar{F}(t, t, t, t). \quad (2)$$

The physical meaning of (2) is as follows:

For a four service point tandem queue with service points of ages s_1, s_2, s_3 and s_4 respectively, equation (2) asserts that the probability that all the four service points are serving t time units from now is the same as if four service points were new.

- vi. If $(X, Y) \sim BVE(\lambda_1, \lambda_2, \lambda_3)$, the covariance between X and Y is given by

$$\text{cov}(X, Y) = \frac{\lambda_3}{(\lambda_1 + \lambda_3)(\lambda_2 + \lambda_3)(\lambda_1 + \lambda_2 + \lambda_3)}, \text{ see Barlow and Proschan (1975).}$$

- vii. The service times T_1, T_2, T_3 and the interarrival time R are independent if and only if $\lambda_4 = 0$.

3. Analysis of the system

Let the symbols 0, 1 and b represent free, busy and blocked states of a service point. Let $X_1(t)$, $X_2(t)$ and $X_3(t)$ denote the states of service point 1, service point 2 and service point 3 at time t respectively. Then, the vector process

$Z(t) = \{(X_1(t), X_2(t), X_3(t)) \mid t \in [0, \infty)\}$ with the state space E given by

$$E = \{(0, 1, 1), (0, b, 1), (1, 1, 1), (1, b, 1), (b, 1, 1), (b, b, 1)\}, \quad (3)$$

denotes the state of the system at time t . Since the quadrivariate exponential distribution proposed by Marshall and Olkin (1967) has exponential marginals and the exponential distribution satisfies LMP, it follows that the stochastic process $\{Z(t)\}$ describing the behaviour of the system is a Markov Process with the infinitesimal generator given by

$$Q = \begin{matrix} & \begin{matrix} (0, 1, 1) & (0, b, 1) & (1, 1, 1) & (1, b, 1) & (b, 1, 1) & (b, b, 1) \end{matrix} \\ \begin{matrix} (0, 1, 1) \\ (0, b, 1) \\ (1, 1, 1) \\ (1, b, 1) \\ (b, 1, 1) \\ (b, b, 1) \end{matrix} & \begin{pmatrix} -(\lambda_1 + \lambda_2 + \lambda_3 + 4\lambda_4) & (\lambda_1 + \lambda_2 + \lambda_4) & (\lambda_3 + 3\lambda_4) & 0 & 0 & 0 \\ 0 & -(\lambda_3 + 3\lambda_4) & 0 & (\lambda_3 + 3\lambda_4) & 0 & 0 \\ 0 & 0 & -2(\lambda_1 + \lambda_2 + \lambda_4) & (\lambda_1 + \lambda_2 + \lambda_4) & (\lambda_1 + \lambda_2 + \lambda_4) & 0 \\ 0 & 0 & 0 & -(\lambda_1 + \lambda_2 + \lambda_4) & 0 & (\lambda_1 + \lambda_2 + \lambda_4) \\ 0 & 0 & 0 & 0 & -(\lambda_1 + \lambda_2 + \lambda_4) & (\lambda_1 + \lambda_2 + \lambda_4) \\ (\lambda_1 + \lambda_2 + \lambda_4) & 0 & 0 & 0 & 0 & -(\lambda_1 + \lambda_2 + \lambda_4) \end{pmatrix} \end{matrix} \quad (4)$$

Let $p_{ijk}(t) = \Pr[Z(t) = (i, j, k)] \forall (i, j, k) \in E$ represent the probability that the system is in state (i, j, k) at time t with the initial condition $p_{011}(0) = 1$. From the infinitesimal generator of the process given in (4), we have the following system of differential-difference equations:

$$\frac{dp_a(t)}{dt} = -(\lambda_1 + \lambda_2 + \lambda_3 + 4\lambda_4)p_a(t) + (\lambda_1 + \lambda_2 + \lambda_4)p_f(t) \quad (5)$$

$$\frac{dp_{b'}(t)}{dt} = (\lambda_1 + \lambda_2 + \lambda_4)p_a(t) - (\lambda_3 + 3\lambda_4)p_{b'}(t) \quad (6)$$

$$\frac{dp_c(t)}{dt} = (\lambda_3 + 3\lambda_4)p_a(t) - 2(\lambda_1 + \lambda_2 + \lambda_4)p_c(t) \quad (7)$$

$$\frac{dp_d(t)}{dt} = (\lambda_3 + 3\lambda_4)p_{b'}(t) + (\lambda_1 + \lambda_2 + \lambda_4)p_c(t) - (\lambda_1 + \lambda_2 + \lambda_4)p_d(t) \quad (8)$$

$$\frac{dp_e(t)}{dt} = (\lambda_1 + \lambda_2 + \lambda_4)p_c(t) - (\lambda_1 + \lambda_2 + \lambda_4)p_e(t) \quad (9)$$

$$\frac{dp_f(t)}{dt} = (\lambda_1 + \lambda_2 + \lambda_4)p_d(t) + (\lambda_1 + \lambda_2 + \lambda_4)p_e(t) - (\lambda_1 + \lambda_2 + \lambda_4)p_f(t) \quad (10)$$

where $a = (0, 1, 1)$, $b' = (0, b, 1)$, $c = (1, 1, 1)$, $d = (1, b, 1)$, $e = (b, 1, 1)$, $f = (b, b, 1)$.

3.1. Transient solution

Solving the system of differential-difference equations given in (5)-(10) along with the equation $\sum_{(i,j,k) \in E} p_{ijk}(t) = 1$ using Laplace Transforms, we get

$$p_a(t) = \frac{2a^3b}{\alpha_1\alpha_2\alpha_3\alpha_4} + \sum_{i=1}^4 \frac{(\alpha_i+2a)(\alpha_i+a)^2(\alpha_i+b)}{\alpha_i \prod_{\substack{j=1 \\ j \neq i}}^4 (\alpha_i - \alpha_j)} e^{\alpha_i t} \quad (11)$$

$$p_{b'}(t) = \frac{2a^4}{\alpha_1\alpha_2\alpha_3\alpha_4} + a \sum_{i=1}^4 \frac{(\alpha_i+2a)(\alpha_i+a)^2}{\alpha_i \prod_{\substack{j=1 \\ j \neq i}}^4 (\alpha_i - \alpha_j)} e^{\alpha_i t} \quad (12)$$

$$p_c(t) = \frac{a^2b^2}{\alpha_1\alpha_2\alpha_3\alpha_4} + b \sum_{i=1}^4 \frac{(\alpha_i+a)^2(\alpha_i+b)}{\alpha_i \prod_{\substack{j=1 \\ j \neq i}}^4 (\alpha_i - \alpha_j)} e^{\alpha_i t} \quad (13)$$

$$p_d(t) = \frac{a^2b(2a+b)}{\alpha_1\alpha_2\alpha_3\alpha_4} + ab \sum_{i=1}^4 \frac{(\alpha_i+a)[2(\alpha_i+a)+b]}{\alpha_i \prod_{\substack{j=1 \\ j \neq i}}^4 (\alpha_i - \alpha_j)} e^{\alpha_i t} \quad (14)$$

$$p_e(t) = \frac{a^2b^2}{\alpha_1\alpha_2\alpha_3\alpha_4} + ab \sum_{i=1}^4 \frac{(\alpha_i+a)(\alpha_i+b)}{\alpha_i \prod_{\substack{j=1 \\ j \neq i}}^4 (\alpha_i - \alpha_j)} e^{\alpha_i t} \quad (15)$$

$$p_f(t) = \frac{2a^2b(a+b)}{\alpha_1\alpha_2\alpha_3\alpha_4} + a^2b \sum_{i=1}^4 \frac{[3\alpha_i+2(a+b)]}{\alpha_i \prod_{\substack{j=1 \\ j \neq i}}^4 (\alpha_i - \alpha_j)} e^{\alpha_i t} \quad (16)$$

where $\alpha_1, \alpha_2, \alpha_3$ and α_4 are the roots of

$$s^4 + (5a+2b)s^3 + (9a^2+9ab+b^2)s^2 + a(7a^2+14ab+4b^2)s + a^2(2a^2+6ab+5b^2) = 0,$$

$$a = (\lambda_1 + \lambda_2 + \lambda_4) \text{ and } b = (\lambda_3 + 3\lambda_4).$$

3.2. Steady state solution

We have seen that the equations (5)–(10) can be solved using the fact that $\sum_{(i,j,k) \in E} p_{ijk}(t) = 1$. Since

we wish to study the stationary behaviour of the system, let $\lim_{t \rightarrow \infty} p_{ijk}(t) = p_{ijk}$. Further, let

$\mathbf{p} = (p_a, p_{b'}, p_c, p_d, p_e, p_f)$ be the stationary distribution corresponding to the Markov process $\{Z(t), t \geq 0\}$. The steady state probabilities are given by

$$p_a = \frac{2(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4)}{[2(\lambda_1 + \lambda_2 + \lambda_4)^2 + 6(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4) + 5(\lambda_3 + 3\lambda_4)^2]} \quad (17)$$

$$p_{b'} = \frac{2(\lambda_1 + \lambda_2 + \lambda_4)^2}{[2(\lambda_1 + \lambda_2 + \lambda_4)^2 + 6(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4) + 5(\lambda_3 + 3\lambda_4)^2]} \quad (18)$$

$$p_c = \frac{(\lambda_3 + 3\lambda_4)^2}{[2(\lambda_1 + \lambda_2 + \lambda_4)^2 + 6(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4) + 5(\lambda_3 + 3\lambda_4)^2]} \quad (19)$$

$$p_d = \frac{(\lambda_3 + 3\lambda_4)[2(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4)]}{[2(\lambda_1 + \lambda_2 + \lambda_4)^2 + 6(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4) + 5(\lambda_3 + 3\lambda_4)^2]} \quad (20)$$

$$p_e = \frac{(\lambda_3 + 3\lambda_4)^2}{[2(\lambda_1 + \lambda_2 + \lambda_4)^2 + 6(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4) + 5(\lambda_3 + 3\lambda_4)^2]} \quad (21)$$

$$p_f = \frac{2(\lambda_3 + 3\lambda_4)(\lambda_1 + \lambda_2 + \lambda_3 + 4\lambda_4)}{[2(\lambda_1 + \lambda_2 + \lambda_4)^2 + 6(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4) + 5(\lambda_3 + 3\lambda_4)^2]}. \quad (22)$$

3.3. Expected number of entities in the system

The expected number of entities in the system is given by

$$\begin{aligned} L_s &= \sum_{n=0}^{\infty} np_n \\ &= 2(p_a + p_{b'}) + 3(p_c + p_d + p_e + p_f) \\ &= \frac{[4(\lambda_1 + \lambda_2 + \lambda_4)^2 + 16(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4) + 15(\lambda_3 + 3\lambda_4)^2]}{[2(\lambda_1 + \lambda_2 + \lambda_4)^2 + 6(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4) + 5(\lambda_3 + 3\lambda_4)^2]} \end{aligned} \quad (23)$$

Particular case:

By choosing $\lambda_4 = 0$, i.e., when the interarrival time and service times at three service points are independent, it is readily seen that

$$L_s = \frac{[4(\lambda_1 + \lambda_2)^2 + 16\lambda_3(\lambda_1 + \lambda_2) + 15\lambda_3^2]}{[2(\lambda_1 + \lambda_2)^2 + 6\lambda_3(\lambda_1 + \lambda_2) + 5\lambda_3^2]}. \quad (24)$$

4. Statistical inference for the applications of quadrivariate exponential distribution to a three service point tandem queue

4.1. Moment estimator for a three service point tandem queue with dependent interarrival time and service times

Let $(Y_{1i}, Y_{2i}, Y_{3i}, Y_{4i})$, $i = 1, 2, \dots, n$ be a random sample size of size n drawn from a quadrivariate exponential interarrival and service times population with the survival function given by (1). It is clear the $\bar{Y}_1, \bar{Y}_2, \bar{Y}_3$ and $\bar{X}$ are the moment estimators of $\frac{1}{\lambda_1 + \lambda_2 + \lambda_4}$, $\frac{1}{\lambda_1 + \lambda_2 + \lambda_4}$, $\frac{1}{\lambda_1 + \lambda_2 + \lambda_4}$ and $\frac{1}{\lambda_3 + 3\lambda_4}$ respectively, where $\bar{Y}_1, \bar{Y}_2, \bar{Y}_3$ and $\bar{X}$ are the sample means of service times at service point 1, service point 2, service point 3 and interarrival times respectively. Let $\theta_1 = \frac{1}{\lambda_1 + \lambda_2 + \lambda_4}$ and $\theta_2 = \frac{1}{\lambda_3 + 3\lambda_4}$. Clearly, the expected number of entities in the system given in (23) is simplified to

$$L_s = \frac{(15\theta_1^2 + 16\theta_1\theta_2 + 4\theta_2^2)}{(5\theta_1^2 + 6\theta_1\theta_2 + 2\theta_2^2)}. \quad (25)$$

Hence, an estimator of L_s based on moments is given by

$$\hat{L}_s = \frac{(15\bar{Y}_1^2 + 16\bar{Y}_1\bar{X} + 4\bar{X}^2)}{(5\bar{Y}_1^2 + 6\bar{X}\bar{Y}_1 + 2\bar{X}^2)}. \quad (26)$$

It may be noted that $\hat{L}_s$ given in (26) is a real valued function in $\bar{X}$ and $\bar{Y}_1$, which is also differentiable.

4.2. Application of Multivariate Central Limit theorem

Suppose $T'_1, T'_2, T'_3, \dots$ are independent and identically distributed k -dimensional random variables such that $T'_n = (T_{1n}, T_{2n}, \dots, T_{kn})$, $n = 1, 2, 3, \dots$ having the first and second order moments $E(T_n) = \mu$ and $\text{var}(T_n) = \Sigma$. Define the sequence of random variables $\bar{T}'_n = (\bar{T}_{1n}, \bar{T}_{2n}, \dots, \bar{T}_{kn})$, $n = 1, 2, 3, \dots$ where $\bar{T}_{in} = \frac{1}{n} \sum_{j=1}^n T_{ij}$, $i = 1, 2, \dots, k$; then $\sqrt{n}(\bar{T}_n - \mu) \xrightarrow{d} N_k(0, \Sigma)$ as $n \rightarrow \infty$.

4.3. CAN Estimator

By applying the multivariate central limit theorem given in section 4.2, it is seen that $\sqrt{n}[(\bar{Y}_1, \bar{Y}_2, \bar{Y}_3, \bar{X}) - (\theta_1, \theta_1, \theta_1, \theta_2)] \xrightarrow{d} N_4(0, \Sigma)$ as $n \rightarrow \infty$, where the dispersion matrix $\Sigma = ((\sigma_{ij}))$ is given by

$$\Sigma = \begin{pmatrix} \bar{Y}_1 & \bar{Y}_2 & \bar{Y}_3 & \bar{X} \\ \bar{Y}_1 & \theta_1^2 & 0 & \frac{\lambda_4}{(\lambda_1 + \lambda_2 + \lambda_3 + 3\lambda_4)(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4)} \\ \bar{Y}_2 & 0 & \theta_1^2 & \frac{\lambda_4}{(\lambda_1 + \lambda_2 + \lambda_3 + 3\lambda_4)(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4)} \\ \bar{Y}_3 & 0 & 0 & \frac{\lambda_4}{(\lambda_1 + \lambda_2 + \lambda_3 + 3\lambda_4)(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4)} \\ \bar{X} & \frac{\lambda_4}{(\lambda_1 + \lambda_2 + \lambda_3 + 3\lambda_4)(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4)} & \frac{\lambda_4}{(\lambda_1 + \lambda_2 + \lambda_3 + 3\lambda_4)(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4)} & \frac{\lambda_4}{(\lambda_1 + \lambda_2 + \lambda_3 + 3\lambda_4)(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4)} & \theta_2^2 \end{pmatrix}. \quad (27)$$

From Rao (1974), we have as $n \rightarrow \infty$

$$\sqrt{n}(\hat{L}_s - L_s) \xrightarrow{d} N_4(0, \sigma^2(\theta)),$$

where $\theta = (\theta_1, \theta_1, \theta_1, \theta_2)$ and

$$\sigma^2(\theta) = 3\theta_1^2 \left(\frac{\partial L_s}{\partial \theta_1} \right)^2 + \theta_2^2 \left(\frac{\partial L_s}{\partial \theta_2} \right)^2 + \frac{6\lambda_4}{(\lambda_1 + \lambda_2 + \lambda_3 + 3\lambda_4)(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4)} \left(\frac{\partial L_s}{\partial \theta_1} \right) \left(\frac{\partial L_s}{\partial \theta_2} \right). \quad (28)$$

It can be shown that the partial derivatives $\left(\frac{\partial L_s}{\partial \theta_1} \right)$ and $\left(\frac{\partial L_s}{\partial \theta_2} \right)$ are given by

$$\left(\frac{\partial L_s}{\partial \theta_1} \right) = \frac{2\theta_2(5\theta_1^2 + 10\theta_1\theta_2 + 4\theta_2^2)}{(5\theta_1^2 + 6\theta_1\theta_2 + 2\theta_2^2)^2} \quad (29)$$

$$\left(\frac{\partial L_s}{\partial \theta_2} \right) = \frac{-2\theta_1(5\theta_1^2 + 10\theta_1\theta_2 + 4\theta_2^2)}{(5\theta_1^2 + 6\theta_1\theta_2 + 2\theta_2^2)^2} \quad (30)$$

Substituting (29) and (30) in (28) and simplifying, it can be shown that

$$\sigma^2(\theta) = \frac{8\theta_1\theta_2(5\theta_1^2 + 10\theta_1\theta_2 + 4\theta_2^2)^2}{(5\theta_1^2 + 6\theta_1\theta_2 + 2\theta_2^2)^4} \left[2\theta_1\theta_2 - \frac{3\lambda_4}{(\lambda_1 + \lambda_2 + \lambda_3 + 3\lambda_4)(\lambda_1 + \lambda_2 + \lambda_4)(\lambda_3 + 3\lambda_4)} \right]. \quad (31)$$

Thus $\hat{L}_s$ is a CAN estimator of L_s . (An estimator T_n is said to be a CAN estimator of $g(\theta)$, if the asymptotic distribution of $\sqrt{n}(T_n - g(\theta))$ is normal). There are several methods for generating CAN estimators of which the Method of Moments and the Method of Maximum Likelihood are commonly used to generate such estimators, see Sinha (1986).

4.4. Confidence Interval for a three service point tandem queue with dependent interarrival time and service times

Let $\sigma^2(\hat{\theta})$ be the estimator of $\sigma^2(\theta)$ obtained by replacing θ by a consistent estimator namely $\hat{\theta} = (\bar{Y}_1, \bar{Y}_2, \bar{Y}_3, \bar{X})$. Let $\hat{\sigma}^2 = \sigma^2(\hat{\theta})$. Since $\sigma^2(\theta)$ is a continuous function of θ , $\hat{\sigma}^2$ is a consistent estimator of $\sigma^2(\theta)$. i.e., $\hat{\sigma}^2 \xrightarrow{P} \sigma^2(\theta)$ as $n \rightarrow \infty$.

By Slutsky's theorem, we have $\sqrt{n} \frac{(\hat{L}_s - L_s)}{\hat{\sigma}} \xrightarrow{d} N(0, 1)$. i.e., $Pr[-k_{\frac{\alpha}{2}} < \frac{\sqrt{n}(\hat{L}_s - L_s)}{\hat{\sigma}} < k_{\frac{\alpha}{2}}] = (1 - \alpha)$, where $k_{\frac{\alpha}{2}}$ is obtained from normal tables. Hence, a $100(1 - \alpha)\%$ asymptotic confidence interval for L_s is given by

$$\hat{L}_s \pm k_{\frac{\alpha}{2}} \frac{\hat{\sigma}}{\sqrt{n}}, \quad (32)$$

where $\hat{\sigma}$ is obtained from

$$\sigma^2(\hat{\theta}) = \frac{8\hat{\theta}_1\hat{\theta}_2(5\hat{\theta}_1^2 + 10\hat{\theta}_1\hat{\theta}_2 + 4\hat{\theta}_2^2)^2}{(5\hat{\theta}_1^2 + 6\hat{\theta}_1\hat{\theta}_2 + 2\hat{\theta}_2^2)^4} \left[2\hat{\theta}_1\hat{\theta}_2 - \frac{-3\lambda_4\hat{\theta}_1^2\hat{\theta}_2^2}{(\hat{\theta}_1 + \hat{\theta}_2 - \lambda_4\hat{\theta}_1\hat{\theta}_2)} \right]. \quad (33)$$

5. Numerical Illustration

To understand the behaviour of the expected number of entities in the system namely L_s as given in (25), a simulation study was carried out by generating samples from two exponential distributions with parameters $(\lambda_1 + \lambda_2 + \lambda_4)$ and $(\lambda_3 + 3\lambda_4)$ respectively using R software version 2.15.0. The choices of the parameters were randomly assigned as $\lambda_1 = 3(0.5)5.5$, $\lambda_2 = 2.5(0.5)5$, $\lambda_3 = 3, 6$ and $\lambda_4 = 1$ respectively. For each choice of the values assigned for the parameters, 1000 samples were generated and the estimated value of expected number of entities in the system ($\hat{L}_s$) given in (26) and the 95% confidence interval for L_s given in (32) were computed. The results obtained are displayed in Tables 2 and 3 respectively.

From Table 2, it is clear that for a fixed value of λ_3 , as $(\lambda_1 + \lambda_2 + \lambda_4)$ increases, the corresponding value of $\hat{L}_s$ decreases. i.e., as the service times at service points 1, 2 and 3 increase, the expected number of entities in the system decreases. However for fixed values of $\lambda_1, \lambda_2, \lambda_4$, if $\lambda_3 + 3\lambda_4$ increases, then the value of $\hat{L}_s$ increases. This means that if the service times at points 1, 2 and 3 remain constant and the interarrival time keeps increasing, then the expected number of entities in the system increases. Similar interpretations hold good for the confidence intervals given in Table 3.

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Table 2: Estimated values of L_5 .

	$\lambda_1 \backslash \lambda_2$	3.0	3.5	4.0	4.5	5.0	5.5
$\lambda_3 = 3$	2.5	2.028709	1.954092	1.884573	1.819671	1.758959	1.702059
	3.0	1.954092	1.884573	1.819671	1.758959	1.702059	1.648635
	3.5	1.884573	1.819671	1.758959	1.702059	1.648635	1.598386
	4.0	1.819671	1.758959	1.702059	1.648635	1.598386	1.551047
	4.5	1.758959	1.702059	1.648635	1.598386	1.551047	1.506376
	5.0	1.702059	1.648635	1.598386	1.551047	1.506376	1.464161
$\lambda_3 = 6$	2.5	2.682845	2.61926	2.558326	2.499911	2.443886	2.390126
	3.0	2.61926	2.558326	2.499911	2.443886	2.390126	2.338515
	3.5	2.558326	2.499911	2.443886	2.390126	2.338515	2.288939
	4.0	2.499911	2.443886	2.390126	2.338515	2.288939	2.241294
	4.5	2.443886	2.390126	2.338515	2.288939	2.241294	2.195479
	5.0	2.390126	2.338515	2.288939	2.241294	2.195479	2.1514

Table 3: 95 % confidence interval for L_5 .

	$\lambda_1 \backslash \lambda_2$	3.0	3.5	4.0	4.5	5.0	5.5
$\lambda_3 = 3$	2.5	(1.999,2.058)	(1.925,1.984)	(1.855,1.914)	(1.790,1.849)	(1.729,1.789)	(1.673,1.732)
	3.0	(1.925,1.984)	(1.855,1.914)	(1.790,1.849)	(1.729,1.789)	(1.673,1.732)	(1.619,1.678)
	3.5	(1.855,1.914)	(1.790,1.849)	(1.729,1.789)	(1.673,1.732)	(1.619,1.678)	(1.569,1.627)
	4.0	(1.790,1.849)	(1.729,1.789)	(1.673,1.732)	(1.619,1.678)	(1.569,1.627)	(1.522,1.580)
	4.5	(1.729,1.789)	(1.673,1.732)	(1.619,1.678)	(1.569,1.627)	(1.522,1.580)	(1.478,1.535)
	5.0	(1.673,1.732)	(1.619,1.678)	(1.569,1.627)	(1.522,1.580)	(1.478,1.535)	(1.436,1.492)
$\lambda_3 = 6$	2.5	(2.657,2.709)	(2.592,2.646)	(2.531,2.586)	(2.472,2.528)	(2.415,2.473)	(2.361,2.419)
	3.0	(2.592,2.646)	(2.531,2.586)	(2.472,2.528)	(2.415,2.473)	(2.361,2.419)	(2.309,2.368)
	3.5	(2.531,2.586)	(2.472,2.528)	(2.415,2.473)	(2.361,2.419)	(2.309,2.368)	(2.259,2.319)
	4.0	(2.472,2.528)	(2.415,2.473)	(2.361,2.419)	(2.309,2.368)	(2.259,2.319)	(2.211,2.272)
	4.5	(2.415,2.473)	(2.361,2.419)	(2.309,2.368)	(2.259,2.319)	(2.211,2.272)	(2.165,2.226)
	5.0	(2.361,2.419)	(2.309,2.368)	(2.259,2.319)	(2.211,2.272)	(2.165,2.226)	(2.121,2.182)

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